

following: 1) *three wins*; 2) two wins, one loss, net *one win*; 3) two wins, one loss, net *one win*; 4) one win, two losses, net *one loss*; 5) two wins, one loss, net *one win*; 6) two losses, one win, net *one loss*; 7) two losses, one win, net *one loss*; and, finally, 8) *three losses*.

Out of the eight cases, the case of three wins occurs once. The case of three losses occurs once. The case of one net loss (one win, two losses) occurs three times. The case of one net win (one loss, two wins) occurs three times.

Play one more round, the fourth. There will be sixteen equally likely outcomes. You will have one case of four wins, one case of four losses, four cases of two wins, four cases of two losses, and six break-even cases.

The quincunx (its name is derived from the Latin for five) in the pinball example shows the fifth round, with thirty-two possibilities, easy to track. Such was the concept behind the quincunx used by Francis Galton. Galton was both insufficiently lazy and a bit too innocent of mathematics; instead of building the contraption, he could have worked with simpler algebra, or perhaps undertaken a thought experiment like this one.

Let's keep playing. Continue until you have forty flips. You can perform them in minutes, but we will need a calculator to work out the number of outcomes, which are taxing to our simple thought method. You will have about 1,099,511,627,776 possible combinations—more than one thousand billion. Don't bother doing the calculation manually, it is two multiplied by itself forty times, since each branch doubles at every juncture. (Recall that we added a win and a lose at the end of the alternatives of the third round to go to the fourth round, thus doubling the number of alternatives.) Of these combinations, only one will be up forty, and only one will be down forty. The rest will hover around the middle, here zero.

We can already see that in this type of randomness extremes are exceedingly rare. One in 1,099,511,627,776 is up forty out of forty tosses. If you perform the exercise of forty flips once per hour, the odds of getting 40 ups in a row are so small that it would take quite a bit of forty-flip trials to see it. Assuming you take a few breaks to eat, argue with your friends and roommates, have a beer, and sleep, you can expect to wait close to four million lifetimes to get a 40-up outcome (or a 40-down outcome) just once. And consider the following. Assume you play one additional round, for a total of 41; to get 41 straight heads would take eight million lifetimes! Going from 40 to 41 halves the odds. This is a key attribute of the nonscalable framework to analyzing randomness: extreme deviations decrease at an increasing rate. You can expect to toss 50 heads in a row once in four billion lifetimes!

FIGURE 9: NUMBERS OF WINS TOSSED